

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

The logs show that: A very large number of TCP SYN packets are being sent from an unfamiliar IP address. The server is receiving more connection requests than it can process, causing it to become overloaded and unable to respond to legitimate users.

This event could be: A **TCP SYN Flood (Denial-of-Service)** attack. The attacker is flooding the server with SYN requests without completing the TCP connection, exhausting the server's resources and preventing legitimate users from accessing the website.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The client sends a **SYN** packet to the server to request a connection.
2. The server replies with a **SYN-ACK** packet to acknowledge the request and indicate it is ready to establish the connection
3. The client responds with an **ACK** packet, completing the three-way handshake and establishing the connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: The attacker sends thousands of SYN packets but never completes the TCP handshake by sending the final ACK packet. The server waits for responses from these fake or incomplete connections, causing many half-open connections to remain active. This consumes server memory and processing resources until legitimate users can no longer establish connections.

Explain what the logs indicate and how that affects the server: The logs indicate an unusually high number of incoming TCP SYN requests from an unfamiliar IP address. This traffic overwhelms the server, preventing it from responding to legitimate connection

requests. As a result, employees and customers experience slow performance, connection timeouts, or are unable to access the company's website. This disruption can affect business operations, reduce customer satisfaction, and potentially result in lost sales.